



12491 - NIRCam and MIRI observations of a young stellar cluster

Cycle: 4, Proposal Category: DD

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Young Stellar Cluster				
	1	StellarCuster-NIRCam	NIRCam Imaging	(1) Cluster-NIRCam
	2	StellarCuster-MIRI	MIRI Imaging	(2) Cluster-MIRI
	3	StellarCuster-MIRI	MIRI Imaging	(3) Cluster-MIRI-subarray
	11	Residual Persistence Check	NIRCam Dark	NONE

ABSTRACT

We will observe this young stellar cluster with 3 NIRCam wide filters, 2 narrow filters, and 1 medium filter, sufficiently deep to allow for the creation of low-noise color-composite images. We will also use the MIRI F770W filter to capture PAHs.

OBSERVING DESCRIPTION

We will observe this young stellar cluster with 3 NIRCам wide filters, 2 narrow filters, and 1 medium filter, sufficiently deep to allow for the creation of low-noise color-composite images. We will also use the MIRI F770W filter to capture PAHs.

Proposal 12491 - Targets - NIRCam and MIRI observations of a young stellar cluster

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	Cluster-NIRCam	RA: 21 43 5.0300 (325.7709583d) Dec: +66 06 45.91 (66.11275d) Equinox: J2000	Proper Motion RA: -1.759 mas/yr Proper Motion Dec: -3.267000033702061 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Young star clusters]				
	(2)	Cluster-MIRI	RA: 21 43 5.0300 (325.7709583d) Dec: +66 06 45.91 (66.11275d) Equinox: J2000	Proper Motion RA: -1.759 mas/yr Proper Motion Dec: -3.267000033702061 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Young star clusters]				
	(3)	Cluster-MIRI-subarray	RA: 21 43 4.4568 (325.7685700d) Dec: +66 06 57.52 (66.11598d) Equinox: J2000	Proper Motion RA: -1.759 mas/yr Proper Motion Dec: -3.267000033702061 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Young star clusters]				

Proposal 12491 - Observation 1 - NIRCcam and MIRI observations of a young stellar cluster

Observation	Proposal 12491, Observation 1: StellarCuster-NIRCam									Wed Dec 03 00:00:28 GMT 2025
	Diagnostic Status: Error									
Observing Template: NIRCam Imaging										
Diagnostics	(StellarCuster-NIRCam (Obs 1)) Error (Form): Permission has not been granted for this program to use value true for field Exclusive Use Of Instrument.									
	(StellarCuster-NIRCam (Obs 1)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.									
	(StellarCuster-NIRCam (Obs 1)) Warning (Form): Data Excess over lower threshold									
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(1)	Cluster-NIRCam	RA: 21 43 5.0300 (325.7709583d) Dec: +66 06 45.91 (66.11275d) Equinox: J2000			Proper Motion RA: -1.759 mas/yr Proper Motion Dec: -3.267000033702061 mas/yr Epoch of Position: 2000				
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Young star clusters]									
	Template	Module		Subarray				Target Placement		
ALL		FULL				Module gap (large extended source)				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)	Tile Order
	3	1	10.0		71.5		0.0		0.0	DEFAULT
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	FULLBOX		6TIGHT		STANDARD				1
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F187N	F470N+F444W	SHALLOW2	5	1	6	6	1417.254	
	2	F115W	F335M	BRIGHT1	7	1	6	6	837.468	
	3	F200W	F444W	BRIGHT1	6	1	6	6	708.627	

Proposal 12491 - Observation 1 - NIRCам and MIRI observations of a young stellar cluster

Special Requirements	Sequence Visits , Non-interruptible, Exclusive Use Of Instrument Aperture PA Range 47 to 50 Degrees (V3 47.07457694 to 50.07457694) Aperture PA Range 54 to 61 Degrees (V3 54.07457694 to 61.07457694) Visits Same PA Persistence Risk
	11 After 1 by 10 Hours to 2 Days, Exclusive Use Of Instrument

Proposal 12491 - Observation 2 - NIRCcam and MIRI observations of a young stellar cluster

Observation	Proposal 12491, Observation 2: StellarCluster-MIRI										Wed Dec 03 00:00:28 GMT 2025
	Diagnostic Status: Warning										
Observing Template: MIRI Imaging											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:10) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:11) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:12) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
(2)		Cluster-MIRI	RA: 21 43 5.0300 (325.7709583d)			Proper Motion RA: -1.759 mas/yr					
			Dec: +66 06 45.91 (66.11275d)			Proper Motion Dec: -3.267000033702061 mas/yr					
			Equinox: J2000			Epoch of Position: 2000					
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Young star clusters]											
Template	Subarray										
	FULL										
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)	Column shift (deg)		Tile Order	
	3	4	10.0		10.0		0.0	0.0		DEFAULT	
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	6						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	5	3	1	Dither 1	6	18	283.054	

Proposal 12491 - Observation 2 - NIRCам and MIRI observations of a young stellar cluster

Special Requirements	Sequence Visits within 53.0 Days Aperture PA Range 45 to 65 Degrees (V3 40.16455103 to 60.16455103) Visits Same PA 3 After 2
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Proposal 12491 - Observation 3 - NIRCcam and MIRI observations of a young stellar cluster

Observation	Proposal 12491, Observation 3: StellarCuster-MIRI										Wed Dec 03 00:00:28 GMT 2025	
	Diagnostic Status: Warning Observing Template: MIRI Imaging											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(3)	Cluster-MIRI-subarray	RA: 21 43 4.4568 (325.7685700d) Dec: +66 06 57.52 (66.11598d) Equinox: J2000				Proper Motion RA: -1.759 mas/yr Proper Motion Dec: -3.267000033702061 mas/yr Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Young star clusters]											
Template	Subarray											
	BRIGHTSKY											
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)	Column shift (deg)		Tile Order		
	1	2	10.0		10.0		0.0	0.0		DEFAULT		
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size		
	1	CYCLING	1	6						DEFAULT		
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	F770W	FASTR1	5	3	1	Dither 1	6	18	88.259		
Special Requirements	Aperture PA Range 45 to 65 Degrees (V3 40.16455103 to 60.16455103)											
	3 After 2											

Proposal 12491 - Observation 11 - NIRCam and MIRI observations of a young stellar cluster

Observation	Proposal 12491, Observation 11: Residual Persistence Check Diagnostic Status: Warning Observing Template: NIRCam Dark							Wed Dec 03 00:00:28 GMT 2025
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
Template	Module	Subarray	Science Template		Occulting Mask		No. of Output Channels	
	ALL	FULL	Imaging				4	
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	MEDIUM8	10	1	1	1	1052.203	
Special Requirements	11 After 1 by 10 Hours to 2 Days, Exclusive Use Of Instrument							